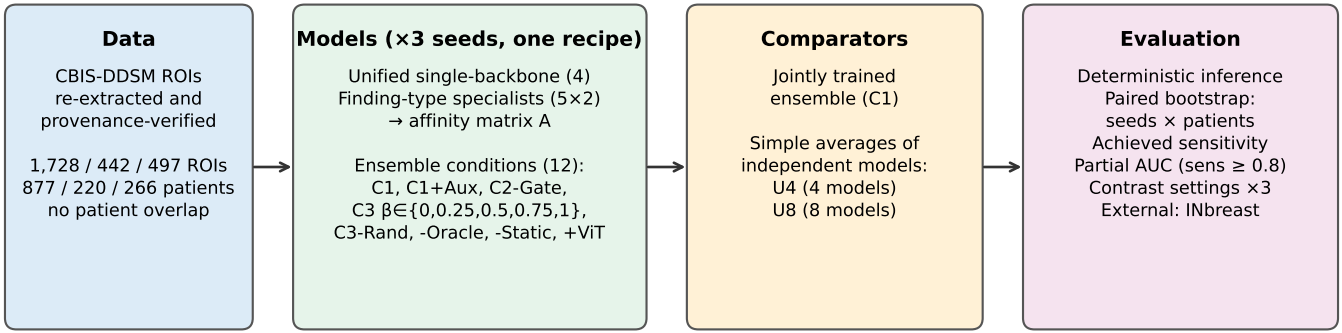
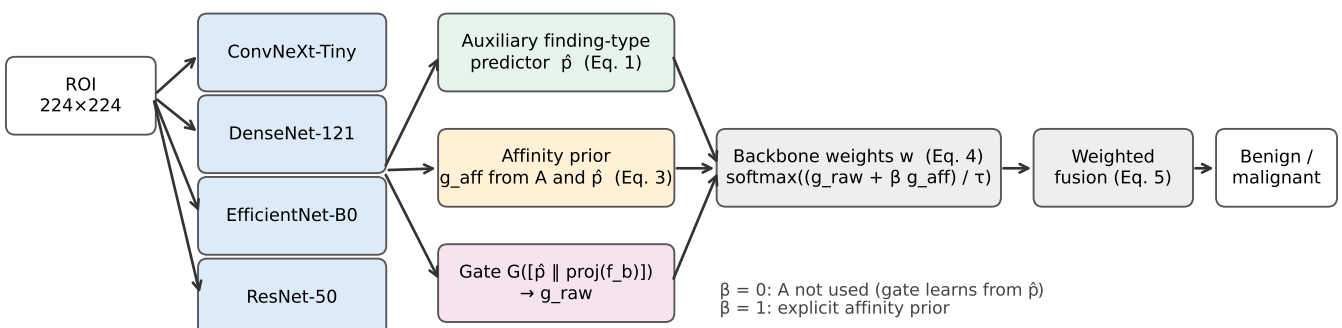

Data
CBIS-DDSM ROIs
re-extracted and
provenance-verified
1,728 / 442 / 497 ROIs
877 / 220 / 266 patients
no patient overlap
Models (×3 seeds, one recipe)
Unified single-backbone (4)
Finding-type specialists (5×2)
→ affinity matrix A
Ensemble conditions (12):
C1, C1+Aux, C2-Gate,
C3 β∈{0,0.25,0.5,0.75,1},
C3-Rand, -Oracle, -Static, +ViT
Comparators
Jointly trained
ensemble (C1)
Simple averages of
independent models:
U4 (4 models)
U8 (8 models)
Evaluation
Deterministic inference
Paired bootstrap:
seeds × patients
Achieved sensitivity
Partial AUC (sens ≥ 0.8)
Contrast settings ×3
External: INbreast

AFAGE (C3)


ROI
224×224
ConvNeXt-Tiny
DenseNet-121
EfficientNet-B0
ResNet-50
Auxiliary finding-type
predictor p̂ (Eq. 1)
Affinity prior
g_aff from A and p̂ (Eq. 3)
Gate G([p̂ ‖ proj(f_b)])
→ g_raw
Backbone weights w (Eq. 4)
softmax((g_raw + β g_aff) / τ)
Weighted
fusion (Eq. 5)
Benign /
malignant
β = 0: A not used (gate learns from p̂)
β = 1: explicit affinity prior